

Math 237A Final

Zih-Yu Hsieh

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Problem 1

Lazarsfeld Problem Set 6 (4):

Let $\mathbb{G} = \mathbb{G}(1, 3)$ be the Grassmannian parametrizing lines in $\mathbb{P}^3$, and fix a line $\ell_0 \subset \mathbb{P}^3$. Let $\Sigma \subset \mathbb{G}$ be the subset of $\mathbb{G}$ corresponding to all lines meeting ℓ_0 , so that (as we've seen before) Σ is a hyperplane section of $\mathbb{G}$.

- Show that Σ has a unique singular point (at the point of $\mathbb{G}$ corresponding to ℓ_0). In fact, Σ is a quadric Q of rank 4 in $\mathbb{P}^4$.
- Let $Q \subset \mathbb{P}^4$ be a quadric of rank 4 in $\mathbb{P}^4$, with singular point $o \in Q$. Let Q' be the proper transform of Q under the blowing up of $o \in \mathbb{P}^4$. Show that Q' is non-singular, and that the fibre of $\nu : Q' \rightarrow Q$ over o is (isomorphic to) $\mathbb{P}^1 \times \mathbb{P}^1$.
- Returning to the realization Σ of Q in (a), define $\Sigma' \subseteq \mathbb{G} \times \ell_0$ to be the incidence correspondance $\Sigma' = \{([\ell], x) \mid x \in \ell \cap \ell_0\}$. Show that Σ' is smooth (non-singular), and that the projection $\mu : \Sigma' \rightarrow \Sigma$ is an isomorphism away from $[\ell_0]$, while $\mu^{-1}[\ell_0]$ is a copy of $\mathbb{P}^1$. This map is called the „small resolution“ of $Q = \Sigma$.

Solution:

- First, recall that the Grassmannian has Plucker Embedding $\mathbb{G}(1, 3) \hookrightarrow \mathbb{P}^5$, where for each line (or a 2-dimensional subspace in k^4), a corresponding matrix $\begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{pmatrix}$ in $\mathbb{P}^5$ corresponds to the point $[X_{12}, X_{13}, X_{14}, X_{23}, X_{24}, X_{34}] \in \mathbb{P}^5$, where each $X_{ij} = \det \begin{pmatrix} x_i & x_j \\ y_i & y_j \end{pmatrix}$.

WLOG, up to a projective linear transform one can assume the point $\ell_0 = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$, which corresponds to the point $[1, 0, 0, 0, 0, 0] \in \mathbb{G}(1, 3) \subset \mathbb{P}^5$. Which, we claim that a line $\ell = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \end{pmatrix} \in \Sigma$ iff $X_{34} = \det \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = 0$.

$\Rightarrow$: If $\ell \in \Sigma$, then it intersects with ℓ_0 . On the level of k^4 , it implies there exists $[s, t] \in \mathbb{P}^1$, such that $s[x_1, x_2, x_3, x_4] + t[y_1, y_2, y_3, y_4] \in \ell_0$, which has the last two coordinates being zero, showing $(sx_3 + ty_3, sx_4 + ty_4) = (s \ t) \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = (0 \ 0)$. With $(s, t) \neq 0$ as a vector, this implies $\det \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = 0$, hence the corresponding Pucker coordinate has $X_{34} = 0$.

$\Leftarrow$: If $X_{34} = \det \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = 0$, there exists nonzero (s, t) , such that $(s \ t) \begin{pmatrix} x_3 & x_4 \\ y_3 & y_4 \end{pmatrix} = (sx_3 + ty_3, sx_4 + ty_4) = 0$. Hence, the vector (in k^4) / point (in $\mathbb{P}^3$) has $[sx_1 + ty_1, sx_2 + ty_2, 0, 0] \in \ell_0$, showing $\ell \cap \ell_0 \neq \emptyset$, or $\ell \in \Sigma$.

Hence, $\Sigma \subset \mathbb{G}(1, 3) \subset \mathbb{P}^5$ is the intersection $\{X_{34} = 0\} \cap \mathbb{G}(1, 3)$; with $\mathbb{G}(1, 3)$ being cut out by the equation $X_{12}X_{34} - X_{13}X_{24} + X_{14}X_{23} = 0$, the defining equation for $\Sigma \subset \mathbb{P}^5$ is $\{X_{34} = 0, X_{14}X_{23} - X_{13}X_{24} = 0\}$. And, notice that since the last coordinate is 0, one can simply project Σ down to $\mathbb{P}^4$ with coordinates $[X_{12}, X_{13}, X_{14}, X_{23}, X_{24}]$, and now the defining equation is $X_{14}X_{23} - X_{13}X_{24} = 0$.

For this equations, the partial derivatives all vanished iff $X_{14} = X_{23} = X_{13} = X_{24} = 0$ (since the partials would precisely result in these for, up to ± 1), showing the only singular

point must be at $[X_{12}, 0, 0, 0, 0] = [1, 0, 0, 0, 0]$. Which, the only line in $\mathbb{P}^3$ with $X_{12} \neq 0$, is ℓ_0 . So, ℓ_0 is the only singular point.

- (b) WLOG, up to change of coordinates, say the singularity occurs at $o = [1, 0, 0, 0, 0]$ (and say $\mathbb{P}^4$ has coordinates $[x_0, x_1, x_2, x_3, x_4]$). Then, if say Q is a quadric defined by homogeneous equation f , then within the affine chart U_{x_0} (where $x_0 \neq 0$, WLOG say $x_0 = 1$), the equation of Q is defined by $f(1, x_1, x_2, x_3, x_4)$, and must have all of its partial derivatives vanished at $x_1 = x_2 = x_3 = x_4 = 0$. This indicates that one can't have any terms with x_0 involved. Which, f can be viewed as a homogeneous polynomial in $x_1, \dots, x_4$.

As a result, when blowing up with respect to this affine chart (say $U_{x_0} \cong \mathbb{A}^4$ with coordinates (x_1, x_2, x_3, x_4)), then one has $\text{Bl}_0 U_{x_0} = \left\{ \begin{pmatrix} x_1 & x_2 & x_3 & x_4 \\ W & X & Y & Z \end{pmatrix} \right\}$ (where $[W, X, Y, Z] \in \mathbb{P}^3$ and the matrix has rank ≤ 1). Which, if pick the affine chart U_W (where $W \neq 0$), then within the blowing-up the proper transform of Q , say Q' , has the equation of Q pulling back to $f(1, x_1, x_1 X, x_1 Y, x_1 Z) = x_1^2 \cdot f(1, 1, X, Y, Z)$ (since f is homogeneous, and for quadric surface the degree is 2), where x_1^2 cuts out the exceptional divisor in this chart, so the defining equation of Q' in U_W is $f(1, 1, X, Y, Z)$. Which, notice that since within $Q \subset \mathbb{P}^4$, the only singularity is o , then with the point $[1, 1, X, Y, Z] \neq o$, hence any partial derivatives with respect to X, Y, Z has f being non-vanishing, showing it's non-singular. If apply similar results, this indicates that it is non-singular at all points.

(Isomorphism to $\mathbb{P}^1 \times \mathbb{P}^1$ now is unclear).

- (c) For $\ell \in \Sigma$ such that $\ell \neq \ell_0$, it's clear that $\ell \cap \ell_0 = \{x\}$ a single point, so there exists a unique $x \in \ell_0$, where $([\ell], x) \in \Sigma'$. Hence, restricting to $\Sigma \setminus \{[\ell_0]\}$, the map $\psi : \Sigma \setminus \{[\ell_0]\} \rightarrow \Sigma'$ has $\psi([\ell]) = ([\ell], \ell \cap \ell_0)$ being a well-defined morphism (since it requires the second coordinate to satisfy the equation for both ℓ and ℓ_0 as a variety in $\mathbb{P}^3$), and it defines an isomorphism onto its image (since each $\ell \neq \ell_0$ corresponds to a unique intersection). Which, the inverse is instead given by the projectio (since $\mu \circ \psi([\ell]) = \mu([\ell], \ell \cap \ell_0) = [\ell]$, while $([\ell], x)$ with $x = \ell \cap \ell_0$ has $\psi \circ \mu([\ell], x) = \psi([\ell]) = ([\ell], x)$).

Finally, the reason why $\mu^{-1}([\ell_0]) = \mathbb{P}^1$, because for any $x \in \ell_0$, one has $x \in \ell_0 \cap \ell_0$, hence $([\ell_0], x) \in \Sigma'$. Which, one has $\mu^{-1}([\ell_0]) = \{([\ell_0], x) \mid x \in \ell_0\}$ has a one-to-one correspondence to ℓ_0 by projecting down to the second coordinates. Then, since $\ell_0 \cong \mathbb{P}^1$ (a copy of $\mathbb{P}^1$), then $\mu^{-1}([\ell_0]) \cong \mathbb{P}^1$.